

Micro-Gesture Recognition of Tongue via Bone Conduction Sound

Shogo Tomaru
The University of Tokyo
Tokyo, Japan
tomaru@akg.t.u-tokyo.ac.jp

Ken Takaki
The University of Tokyo
Tokyo, Japan
takaki@akg.t.u-tokyo.ac.jp

Hiroaki Murakami
The University of Tokyo
Tokyo, Japan
murakami@akg.t.u-tokyo.ac.jp

Damyon Kim
The University of Tokyo
Tokyo, Japan
kim_d@akg.t.u-tokyo.ac.jp

Koya Narumi
The University of Tokyo
Tokyo, Japan
narumi@akg.t.u-tokyo.ac.jp

Mitsuhiro Kamezaki
The University of Tokyo
Tokyo, Japan
kamezaki@akg.t.u-tokyo.ac.jp

Yoshihiro Kawahara
The University of Tokyo
Tokyo, Japan
kawahara@akg.t.u-tokyo.ac.jp

ABSTRACT

We propose a hands-free and less perceptible gesture sensing method of the tongue by capturing the bone conduction sound generated when the tongue rubs the teeth. The sound is captured by the bone conduction microphones attached behind the ears. In this work, we show that tongue slide, snap, and teeth click gestures can be classified using the decision tree algorithm, which focuses on the characteristics in the sound spectrogram. We conducted a preliminary experiment to verify that input methods for mouth microgesture devices using bone conduction can be expanded from only teeth to teeth and tongue gestures without any additional obtrusive sensors. The evaluation revealed that our method achieved a classification accuracy of 82.7% with user-specific parameter adjustment.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile computing systems and tools; Sound-based input / output; Gestural input.

KEYWORDS

wearable sensing, mouth gesture recognition, bone conduction, teeth/tongue input

ACM Reference Format:

Shogo Tomaru, Ken Takaki, Hiroaki Murakami, Damyon Kim, Koya Narumi, Mitsuhiro Kamezaki, and Yoshihiro Kawahara. 2024. Micro-Gesture Recognition of Tongue via Bone Conduction Sound. In *The 37th Annual ACM Symposium on User Interface Software and Technology (UIST Adjunct '24)*, October 13–16, 2024, Pittsburgh, PA, USA. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3672539.3686336>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

UIST Adjunct '24, October 13–16, 2024, Pittsburgh, PA, USA

© 2024 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-0718-6/24/10

<https://doi.org/10.1145/3672539.3686336>

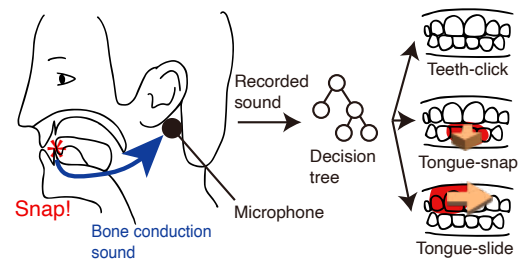


Figure 1: Our method classifies three types of mouth micro-gestures, including tongue gestures. Clenching the teeth and snapping/sliding the tongue against the teeth creates bone conduction sound measurable by bone conduction microphones placed behind the ears.

1 INTRODUCTION AND RELATED WORK

As wearable devices such as smart glasses and wireless earphones become more widespread, opportunities to input commands while hands are occupied (e.g., operating a machine or carrying objects) are increasing. However, we mostly interact with wearable devices using our hands, and such interactions are difficult in the aforementioned scenarios. This issue also presents a significant challenge for individuals with hand disabilities or injuries.

Voice and gaze input techniques are prevalent in current wearable devices. Voice input can be inappropriate in certain social contexts as a risk exposure of private information [5], and gaze input interactions impose a high physical and mental load and exhibit low target selection accuracy [4].

Mouth-based gestures are good alternatives with less privacy concerns and a low physical and mental load. For example, silent speech techniques [3] can recognize speech commands without audible sounds, although conversation contents may leak from the visible mouth movements. Teeth gestures done with minimal mouth movements offer more imperceptible inputs. Yet, they are limited in pattern diversity due to restricted movement of the lower jaw [8].

Tongue gestures are promising for hands-free and eyes-free interactions, as they impose less effort and provide more gesture patterns than teeth gestures [1]. However, current recognition techniques are intrusive, often requiring sensors inside the mouth [6, 9, 10], or around the mouth, such as electroencephalography (EEG), electromyography (EMG), and skin deformation sensing [2, 7, 12], leading to user's discomfort unsuitable for prolonged use.

This work proposes a hands-free and less perceptible gesture sensing method of the tongue using the sound generated by a tongue rubbing the teeth (Figure 1). Tongue movements, which are usually challenging to detect, are captured as bone conduction sounds by the microphones attached behind the ears (i.e., far from the mouth) without any additional obtrusive sensors. To show that our method expands the variety of mouth gestures that could be sensed by inobtrusive bone conduction microphones from only teeth to tongues, we classify teeth click, tongue snap, and tongue slide gestures (Figure 1) based on sound spectrogram. Our preliminary evaluation with four participants revealed that the three types of gestures could be classified with an accuracy of 82.7% with a user-specific parameter adjustment. We also plan to introduce location sensing and single and double tap gestures in future work, which can increase gestures to 18 patterns.

2 MOUTH GESTURE RECOGNITION VIA BONE CONDUCTION SOUND

Our key observation is that snapping or sliding the tongue against the teeth causes a bone conduction sound propagating through the bone, which can be measured behind the ear as the teeth-click sound. Figure 2 (a), (b), (c) shows the frequency domain signal of the teeth and tongue gestures. We found that all gestures exhibit pulses with frequency components within the 100 Hz to 750 Hz range. Regarding the teeth-click gesture, it also has significant high-frequency components within the 4000 Hz to 6000 Hz range, setting it from the tongue gestures. The two tongue gestures, tongue-snap and tongue-slide, do not have distinct differences in the frequency domain but differ significantly in the time domain. The tongue-slide is distinguished by its unique pattern of consecutive pulses.

To classify teeth and tongue gestures, we first determine peaks exhibiting energy above a certain level (as defined in the preliminary study) within the 100–750 Hz range as gesture candidates. Then, peaks having intense energy in both the 100–750 Hz and 4000–6000 Hz ranges are classified as teeth-click gestures. Other gestures are classified as tongue gestures. When multiple peaks are detected within a 1 s time frame, they are classified as a single tongue-slide gesture. Otherwise, they are classified as tongue-snap gestures.

3 EXPERIMENT AND RESULTS

User evaluation procedure. Four authors participated in the experiment conducted inside a quiet room. Each participant performed one practice session and one actual session at one location. During each session, the participants repeated the same gesture fifteen times at four-second intervals with a natural intensity. We used four positions for teeth-click (the canine teeth on both sides, the central incisor teeth, and the molar teeth), three positions for tongue-snap (snapping three regions of the lower teeth with the tongue: left, right, and front), and two positions for tongue-slide (sliding the

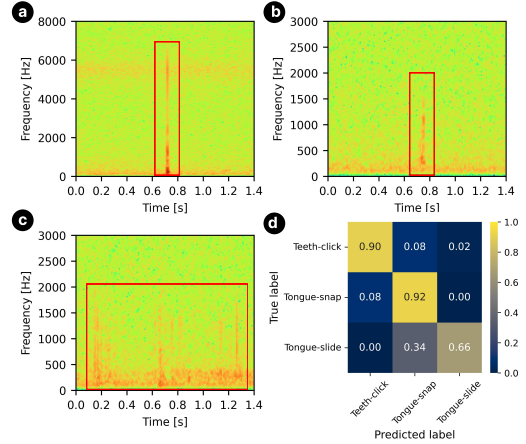


Figure 2: Spectrogram of teeth and tongue gestures for (a) teeth-click, (b) tongue-snap, (c) tongue-slide, and (d) the confusion matrix of the three types of mouth micro-gestures.

inner surface of the upper teeth from the left to right or the opposite direction) to demonstrate that tongue movements can be distinguished from teeth movements regardless of the location where the gestures were performed. Thus, 60, 45, and 30 gestures were collected for each type, respectively. The bone conduction microphone was positioned on the head, between the mastoid and the Mandibular Condyle, based on the observation in Byte.it [11] that this position can record teeth-born sounds efficiently.

Classification results. The energy thresholds used in the algorithm are determined for each participant from the training data, split from the measurement data in a 3 : 2 ratio for training and test. Figure 2 (d) shows the confusion matrix of the classification results for four authors. Teeth-click and tongue-snap were classified with an accuracy exceeding 90%. However, the classification accuracy of tongue-slide was 66%, and 34% of the tongue-slide gestures were misclassified as tongue snaps.

4 CONCLUSION AND FUTURE WORK

We proposed classifying three types of mouth micro-gestures using a bone conduction microphone behind the ear, including teeth-click, tongue-snap, and tongue-slide. We solved the challenge of sensing subtle tongue gestures with unobtrusive sensors by measuring the bone conduction sound of the tongue rubbing the teeth and the decision-tree algorithm. In future work, we plan to increase the number of gestures to 18 by introducing location sensing (left, middle, and right) and single and double gestures and validate the robustness of classification under daily life scenarios.

ACKNOWLEDGMENTS

This work was supported by JSPS Grant-in-Aid for JSPS Fellows Grant Number JP21J22274 and Value Exchange Engineering, a joint research project between R4D, Mercari Inc., and the RIISE.

REFERENCES

- [1] Victor Chen, Xuhai Xu, Richard Li, Yuanchun Shi, Shwetak Patel, and Yuntao Wang. 2021. Understanding the Design Space of Mouth Microgestures. In *Proceedings of the 2021 ACM Designing Interactive Systems Conference* (Virtual Event, USA) (DIS '21). Association for Computing Machinery, New York, NY, USA, 1068–1081. <https://doi.org/10.1145/3461778.3462004>
- [2] Jingyuan Cheng, Ayano Okoso, Kai Kunze, Niels Henze, Albrecht Schmidt, Paul Lukowicz, and Koichi Kise. 2014. On the Tip of My Tongue: A Non-Invasive Pressure-Based Tongue Interface. In *Proceedings of the 5th Augmented Human International Conference* (Kobe, Japan) (AH '14). Association for Computing Machinery, New York, NY, USA, Article 12, 4 pages. <https://doi.org/10.1145/2582051.2582063>
- [3] B. Denby, T. Schultz, K. Honda, T. Hueber, J.M. Gilbert, and J.S. Brumberg. 2010. Silent speech interfaces. *Speech Communication* 52, 4 (2010), 270–287. <https://doi.org/10.1016/j.specom.2009.08.002>
- [4] Morten Lund Dybdal, Javier San Agustin, and John Paulin Hansen. 2012. Gaze Input for Mobile Devices by Dwell and Gestures. In *Proceedings of the Symposium on Eye Tracking Research and Applications* (Santa Barbara, California) (ETRA '12). Association for Computing Machinery, New York, NY, USA, 225–228. <https://doi.org/10.1145/2168556.2168601>
- [5] Susumu Harada, James A. Landay, Jonathan Malkin, Xiao Li, and Jeff A. Bilmes. 2006. The Vocal Joystick: Evaluation of Voice-Based Cursor Control Techniques. In *Proceedings of the 8th International ACM SIGACCESS Conference on Computers and Accessibility* (Portland, Oregon, USA) (Assets '06). Association for Computing Machinery, New York, NY, USA, 197–204. <https://doi.org/10.1145/1168987.1169021>
- [6] Richard Li, Jason Wu, and Thad Starner. 2019. TongueBoard: An Oral Interface for Subtle Input. In *Proceedings of the 10th Augmented Human International Conference 2019* (Reims, France) (AH2019). Association for Computing Machinery, New York, NY, USA, Article 1, 9 pages. <https://doi.org/10.1145/3311823.3311831>
- [7] Phuc Nguyen, Nam Bui, Anh Nguyen, Hoang Truong, Abhijit Suresh, Matt Whitlock, Duy Pham, Thang Dinh, and Tam Vu. 2018. TYTH-Typing On Your Teeth: Tongue-Teeth Localization for Human-Computer Interface. In *Proceedings of the 16th Annual International Conference on Mobile Systems, Applications, and Services* (Munich, Germany) (MobiSys '18). Association for Computing Machinery, New York, NY, USA, 269–282. <https://doi.org/10.1145/3210240.3210322>
- [8] Jay Prakash, Zhiqian Yang, Yu-Lin Wei, Haitham Hassanieh, and Romit Roy Choudhury. 2020. EarSense: Earphones as a Teeth Activity Sensor. In *Proceedings of the 26th Annual International Conference on Mobile Computing and Networking* (London, United Kingdom) (MobiCom '20). Association for Computing Machinery, New York, NY, USA, Article 40, 13 pages. <https://doi.org/10.1145/3372224.3419197>
- [9] Himanshu Sahni, Abdelkareem Bedri, Gabriel Reyes, Pavleen Thukral, Zehua Guo, Thad Starner, and Maysam Ghovanloo. 2014. The Tongue and Ear Interface: A Wearable System for Silent Speech Recognition. In *Proceedings of the 2014 ACM International Symposium on Wearable Computers* (Seattle, Washington) (ISWC '14). Association for Computing Machinery, New York, NY, USA, 47–54. <https://doi.org/10.1145/2634317.2634322>
- [10] T. Scott Saponas, Daniel Kelly, Babak A. Parviz, and Desney S. Tan. 2009. Optically Sensing Tongue Gestures for Computer Input. In *Proceedings of the 22nd Annual ACM Symposium on User Interface Software and Technology* (Victoria, BC, Canada) (UIST '09). Association for Computing Machinery, New York, NY, USA, 177–180. <https://doi.org/10.1145/1622176.1622209>
- [11] Tomás Vega Gálvez, Shardul Sapkota, Alexandru Dancu, and Pattie Maes. 2019. ByteIt: Discreet Teeth Gestures for Mobile Device Interaction. In *Extended Abstracts of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland Uk) (CHI EA '19). Association for Computing Machinery, New York, NY, USA, 1–6. <https://doi.org/10.1145/3290607.3312925>
- [12] Qiao Zhang, Shyamnath Gollakota, Ben Taskar, and Raj P.N. Rao. 2014. Non-Intrusive Tongue Machine Interface. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Toronto, Ontario, Canada) (CHI '14). Association for Computing Machinery, New York, NY, USA, 2555–2558. <https://doi.org/10.1145/2556288.2556981>